

RIVER BEND NUCLEAR GENERATING STATION

(Fictional Facility — Portfolio Demonstration Only)

PERSONAL PROTECTIVE EQUIPMENT (PPE) PLAN

Document Number: RBS-SAF-4410

Revision: 3

Effective Date: March 15, 2025

SAMPLE PORTFOLIO DOCUMENT — NOT FOR OPERATIONAL USE

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1.0 DOCUMENT CONTROL

Field	Detail
Document Number	RBS-SAF-4410
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Role	Name / Title	Signature	Date
Prepared By	M. Langley, Technical Writer — Technical Publications		
Reviewed By	R. Thompson, Radiation Protection Manager		
Approved By	D. Whitfield, Plant Manager		

1.1 Revision History

Rev	Date	Description of Change	Prepared By	Approved By
0	03/01/2019	Initial issue.	J. Alvarez	R. Thompson
1	11/15/2020	Updated respiratory protection requirements to align with revised 10 CFR 20 guidance.	J. Alvarez	R. Thompson
2	02/10/2023	Added arc flash PPE requirements per updated NFPA 70E edition; revised switchyard table.	M. Langley	R. Thompson
3	03/15/2025	Biennial review. Updated hazard matrix, added Human Performance Tools section, revised appendices.	M. Langley	D. Whitfield

2.0 PURPOSE

This procedure establishes the requirements for the selection, use, inspection, maintenance, and storage of Personal Protective Equipment (PPE) at River Bend Nuclear Generating Station (RBS). It defines minimum and task-specific PPE requirements by work area, describes donning and doffing practices that minimize the spread of contamination, and assigns responsibility for PPE program implementation across plant organizations.

This plan supports the station's commitment to maintaining exposures and industrial hazards As Low As Reasonably Achievable (ALARA) and to complying with applicable federal regulations, consensus standards, and station administrative procedures.

3.0 SCOPE

This procedure applies to all RBS employees, contractors, vendors, and visitors who enter station work areas, including the Turbine Building, Reactor Building, Auxiliary Building, Radwaste Area, Switchyard, Chemical Storage areas, Maintenance Shops, Control Room, and any area posted as Contaminated or a High Radiation Area.

This procedure does not supersede site-specific Radiation Work Permits (RWPs), Job Hazard Analyses (JHAs), or Confined Space Entry Permits, which may impose PPE requirements more restrictive than the baseline requirements described herein. Where a conflict exists, the more restrictive requirement governs.

4.0 REFERENCES

4.1 Regulatory Requirements

- 10 CFR 20 — Standards for Protection Against Radiation
- 10 CFR 50, Appendix B — Quality Assurance Criteria for Nuclear Power Plants
- 29 CFR 1910.132 — General Requirements for Personal Protective Equipment
- 29 CFR 1910.133 — Eye and Face Protection
- 29 CFR 1910.134 — Respiratory Protection
- 29 CFR 1910.135 — Head Protection
- 29 CFR 1910.136 — Foot Protection
- 29 CFR 1910.137 — Electrical Protective Equipment
- 29 CFR 1910.138 — Hand Protection
- 29 CFR 1910.146 — Permit-Required Confined Spaces
- 29 CFR 1910.120 — Hazardous Waste Operations and Emergency Response (HAZWOPER)

4.2 Consensus Standards

- NFPA 70E — Standard for Electrical Safety in the Workplace
- ANSI/ISEA Z89.1 — American National Standard for Industrial Head Protection
- ANSI Z87.1 — Occupational and Educational Eye and Face Protection Devices
- ANSI/ISEA 105 — Hand Protection Selection Criteria
- ANSI/ASSP Z359 — Fall Protection Code
- ASTM F2413 — Standard Specification for Performance Requirements for Protective Footwear

4.3 NRC Guidance

- Regulatory Guide 8.15 — Acceptable Programs for Respiratory Protection
- NUREG-0041 — Manual of Respiratory Protection Against Airborne Radioactive Material

4.4 Station Administrative Procedures

- RBS-ADM-1001 — Radiation Protection Program
- RBS-ADM-2015 — Personnel Contamination Control
- RBS-HP-3002 — Access Control to Radiologically Controlled Areas
- RBS-SAF-1100 — Industrial Safety Program
- RBS-SAF-4415 — Respiratory Protection Program
- RBS-SAF-4420 — Fall Protection Program

5.0 DEFINITIONS

Term	Definition
Anti-Contamination Clothing (Anti-Cs)	Protective clothing worn to prevent the spread of, or personal contamination by, radioactive material. Not a shielding device.
Contamination	The presence of radioactive material in an unwanted location, including on PPE, surfaces, or skin.
Dosimeter	A device worn by an individual to measure and record radiation dose. Includes electronic dosimeters (EDs) and thermoluminescent dosimeters (TLDs).
High Radiation Area	An area where radiation levels could result in an individual receiving a dose in excess of 0.1 rem (100 mrem) in one hour at 30 cm from the source.

Term	Definition
Locked High Radiation Area	A high radiation area with radiation levels exceeding 1,000 mrem/hr, secured to prevent unauthorized entry per 10 CFR 20.
Radiation Work Permit (RWP)	A controlled document issued by Radiation Protection authorizing entry into a radiologically controlled area and specifying required PPE, dosimetry, and stay times.
Radiologically Controlled Area (RCA)	An area under the administrative control of the Radiation Protection organization due to actual or potential radiological hazards.
Respirator, Air-Purifying (APR)	A respirator that removes contaminants from ambient air using filters, cartridges, or canisters.
Self-Contained Breathing Apparatus (SCBA)	An atmosphere-supplying respirator providing an independent air supply, used in IDLH or oxygen-deficient atmospheres.
Step-Off Pad	A designated boundary at the exit of a contaminated area where outer PPE is removed and contamination surveys are performed.
Stay Time	The maximum authorized duration an individual may remain in a radiological area based on dose rate and administrative dose limits.
Arc Flash Boundary	The distance from exposed energized electrical conductors within which a person could receive a second-degree burn if an electrical arc flash were to occur.

6.0 RESPONSIBILITIES

6.1 Plant Manager

- Provides organizational resources necessary to implement and maintain the PPE program.
- Ensures PPE performance is reviewed as part of the station safety and radiation protection performance indicators.

6.2 Radiation Protection Manager

- Establishes PPE requirements for radiologically controlled areas and specifies requirements on RWPs.
- Approves respiratory protection use for airborne radioactivity and maintains the site Respiratory Protection Program.
- Ensures contamination surveys, frisking stations, and step-off pads are maintained and functional.

6.3 Operations

- Verifies PPE postings and area boundaries are maintained and communicated during shift turnover.
- Ensures operators don and use PPE appropriate to posted area hazards prior to entry.

6.4 Maintenance

- Performs Job Hazard Analyses prior to work and selects PPE appropriate to identified hazards.
- Ensures tools and work practices are compatible with required PPE (e.g., arc-rated clothing, cut-resistant gloves).

6.5 Contractors

- Complies with RBS PPE requirements as a condition of site access, including area-specific and task-specific PPE.
- Provides PPE meeting or exceeding station and regulatory standards unless station-issued PPE is specified.

6.6 Employees

- Wear PPE as specified for the work area and task, and use stop-work authority if PPE is damaged, unavailable, or inadequate for the hazard.
- Inspect PPE before each use and report deficiencies to their supervisor or the Safety Department.

6.7 Safety Department

- Maintains the station Hazard Matrix and Job Hazard Assessment program.

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- Conducts respirator fit testing, PPE training, and periodic field observations of PPE use.
 - Investigates PPE-related events and tracks corrective actions to completion.

7.0 REQUIRED PPE BY WORK AREA

The following tables identify baseline and task-specific PPE by work area. Baseline PPE is required for entry regardless of task; task-specific PPE is required when the associated hazard is present or specified on the governing RWP, JHA, or work order. Area postings and RWP requirements always govern over this table when more restrictive.

7.1 Turbine Building

PPE Item	Requirement	Standard / Notes
Hard hat	Required at all times	ANSI/ISEA Z89.1 Type I
Safety glasses	Required at all times	ANSI Z87.1
Hearing protection	Required in posted high-noise areas	NRR $\geq$ 25 dB; dual protection above 100 dBA
Steel-toe boots	Required at all times	ASTM F2413
Hearing protection (dual)	Task-specific	Areas with turbine/generator in operation
Cut-resistant gloves	Task-specific	Sheet metal, lagging, and insulation work

7.2 Reactor Building

PPE Item	Requirement	Standard / Notes
Hard hat, safety glasses, steel-toe boots	Required at all times	Baseline PPE
Anti-contamination clothing	Required per RWP	RBS-ADM-2015
Electronic dosimeter	Required for all RCA entries	Alarm set points per RWP
Respirator (APR or PAPR)	Task-specific	Per RWP and ALARA review; RBS-SAF-4415
Hearing protection	Required near operating equipment	NRR $\geq$ 25 dB

7.3 Auxiliary Building

PPE Item	Requirement	Standard / Notes
Hard hat, safety glasses, steel-toe boots	Required at all times	Baseline PPE
Anti-contamination clothing	Required in posted RCA	Per RWP
Electronic dosimeter	Required in RCA	Per RWP
Chemical-resistant gloves	Task-specific	Sampling and chemistry-related tasks
Face shield	Task-specific	Sampling of process fluids

7.4 Radwaste Area

PPE Item	Requirement	Standard / Notes
Anti-contamination clothing (full)	Required at all times	Coveralls, shoe covers, hood as specified on RWP
Double gloving (inner/outer)	Required at all times	Outer glove changed per contamination survey
Electronic dosimeter	Required at all times	Alarming dosimeter per RWP
Full-face respirator or PAPR	Task-specific	Waste handling and resin transfer operations
Shoe covers (booties)	Required at all times	Removed at step-off pad only

7.5 Switchyard

PPE Item	Requirement	Standard / Notes
Hard hat, safety glasses, steel-toe boots	Required at all times	Baseline PPE
Arc flash suit	Task-specific	Per NFPA 70E incident energy analysis and arc flash label
Electrical gloves (rated)	Task-specific	Class per voltage; tested per ASTM D120
Face shield / arc-rated hood	Task-specific	Required within arc flash boundary
Dielectric boots	Task-specific	Switching operations on energized equipment

7.6 Chemical Storage

PPE Item	Requirement	Standard / Notes
Chemical splash goggles	Required at all times	ANSI Z87.1 splash-rated
Chemical-resistant suit	Task-specific	Per SDS for chemical handled
Chemical-resistant gloves	Required for handling	Selected per ANSI/ISEA 105 and SDS compatibility chart
Face shield	Task-specific	Transfer and mixing operations
Air-purifying respirator (cartridge per SDS)	Task-specific	Per exposure assessment; RBS-SAF-4415

7.7 Maintenance Shops

PPE Item	Requirement	Standard / Notes
Hard hat, safety glasses, steel-toe boots	Required at all times	Baseline PPE
Hearing protection	Required at posted machinery	NRR ≥ 25 dB
Face shield	Task-specific	Grinding, buffing, machining
Cut-resistant gloves	Task-specific	Sheet metal and fabrication work
Welding helmet and leathers	Task-specific	Welding and cutting operations

7.8 Control Room

PPE Item	Requirement	Standard / Notes
Station-issued footwear (safety-toe optional)	Per local posting	Control Room is a low-hazard administrative area
Hearing protection	Not routinely required	Alarm audibility must not be impaired
Anti-contamination clothing	Not required	Control Room is outside the RCA boundary

7.9 Contaminated Areas

PPE Item	Requirement	Standard / Notes
Anti-contamination clothing (full)	Required at all times	Per RWP; inspected before donning
Double gloving with taping	Required at all times	Outer glove/sleeve interface taped
Shoe covers	Required at all times	Two pair where posted
Electronic dosimeter	Required at all times	Verified operational before entry
PAPR or full-face respirator	Task-specific	Per airborne radioactivity survey results

7.10 High Radiation Areas

PPE Item	Requirement	Standard / Notes
All Contaminated Area PPE (Section 7.9)	Required as applicable	Per posted area classification
Alarming electronic dosimeter	Required at all times	Audible and vibration alarm enabled
Extremity dosimeter (ring badge)	Task-specific	Required for hands-on work near high dose-rate sources
RWP pre-job briefing	Required before entry	Stay time and dose budget communicated per RBS-HP-3002

8.0 DONNING PROCEDURES

8.1 General Anti-Contamination Clothing Donning Sequence

1. Review the RWP and verify PPE requirements match the planned task and area classification.
2. Inspect all PPE items per Section 10 before donning. Do not don damaged or out-of-date PPE.
3. Remove personal items (jewelry, wallets, phones) and secure them outside the RCA per station policy.
4. Don inner surgical-style gloves.
5. Don anti-contamination coveralls, ensuring cuffs remain accessible for glove attachment.
6. Don shoe covers (booties) over station-issued or personal footwear.
7. Don outer gloves over the coverall cuff and tape the glove-to-sleeve interface.
8. If a respirator is required, perform donning per Section 8.2 prior to entry.
9. Obtain and verify operation of the electronic dosimeter; confirm alarm set points with Radiation Protection.
10. Perform a peer check (Section 15.2) with another qualified individual to verify complete and correct donning.
11. Proceed to the RCA entry point and complete RWP sign-in.

8.2 Respirator Donning and Fit Check

12. Inspect the respirator per Section 10.2 and confirm the correct cartridge or filter is installed for the hazard.
13. Position the respirator on the face and pull straps over the head, seating the lower strap first.
14. Adjust straps evenly until the respirator seals against the face without over-tightening.
15. Perform a positive-pressure seal check by covering the exhalation valve and exhaling gently.
16. Perform a negative-pressure seal check by covering the inlet(s) and inhaling gently; the facepiece should collapse slightly.
17. Report any inability to achieve a proper seal to the Radiation Protection or Safety Department immediately; do not proceed with entry.

9.0 DOFFING PROCEDURES

Doffing sequence is designed to prevent the transfer of contamination from outer PPE surfaces to skin, clothing, or unrestricted areas. Doffing is performed at the designated step-off pad unless otherwise directed by Radiation Protection.

18. Approach the step-off pad and stop for a frisk survey of gloves and visible PPE surfaces before proceeding.
19. Remove tape from the glove-to-sleeve interface without touching the skin or inner glove.
20. Remove outer gloves by rolling them inside-out, avoiding contact between the outer surface and skin.
21. Perform a frisk survey of the coverall surface; if contamination is detected, follow RBS-ADM-2015 for extended decontamination.
22. Unzip and remove the coverall by rolling it inward from the shoulders down, keeping the outer surface contained.
23. Step out of the coverall onto the clean side of the step-off pad without allowing footwear to contact the contaminated side.
24. Remove shoe covers last, stepping directly onto the clean floor area.
25. Remove inner gloves by the cuff, turning them inside-out, and dispose of them in the designated radioactive waste receptacle.
26. Perform a whole-body frisk survey per RBS-ADM-2015 before exiting the RCA.
27. Return the electronic dosimeter to the reader/charging station and record the dose received for the entry.
28. Wash hands and face at the designated hygiene station before eating, drinking, or leaving the controlled area.

Contamination Prevention Note: Never touch the face, hair, or unprotected skin while wearing outer PPE. If contamination is suspected at any point during doffing, stop and notify Radiation Protection before continuing.

10.0 INSPECTION REQUIREMENTS

PPE is inspected before each use. Items that fail inspection are removed from service per Section 13.

PPE Type	Inspection Criteria	Frequency
Hard hat	Shell free of cracks, dents, or UV degradation; suspension intact; manufacture date within service life	Before each use
Respirator	Facepiece, valves, and straps intact; cartridge within service life; seal check performed	Before each use
Anti-contamination clothing	Free of tears, holes, and prior unremoved contamination tags	Before each use
Electronic dosimeter	Powers on, reads zero (or background), alarm function tested	Before each entry
Fall protection harness	Webbing free of cuts/fraying, hardware free of cracks/corrosion, stitching intact	Before each use and annually by competent person
Electrical gloves	Visual inspection and air (inflation) test for punctures; within retest date	Before each use
Arc flash suit	Free of holes, contamination, or damaged reflective layers; within manufacturer service life	Before each use

11.0 CLEANING AND DECONTAMINATION

- Anti-contamination clothing is laundered only through the station's contracted radiological laundry service; it is never laundered off site or in station domestic laundry facilities.
- Reusable respirators are decontaminated per manufacturer instructions and RBS-SAF-4415 after each use, and inspected before being returned to service.
- Non-porous PPE (hard hats, face shields, goggles) is wiped down with approved cleaning solution between users.
- Chemical-resistant PPE is decontaminated per the applicable Safety Data Sheet (SDS) for the chemical encountered.
- PPE that cannot be adequately decontaminated is disposed of as radioactive waste or hazardous waste, as applicable, per station waste procedures.

12.0 STORAGE REQUIREMENTS

- Clean PPE is stored separately from PPE used in radiologically controlled or contaminated areas.
- Respirators are stored in sealed containers or bags, away from direct sunlight, extreme temperature, and chemical vapors.
- Fall protection equipment is stored away from UV exposure, moisture, and chemical contact per manufacturer guidance.
- Electronic dosimeters are stored in the designated charging/reader station when not in use.
- Arc flash suits are stored in a garment bag or cabinet to protect the reflective and insulating layers from abrasion.

13.0 PPE REPLACEMENT CRITERIA

PPE Type	Replacement Trigger
Any PPE item	Visible damage, failed inspection, or missing certification/date markings
Respirator cartridges/filters	Reached end-of-service-life indicator, expiration date, or change-out schedule
Fall protection harness/lanyard	Involved in a fall-arrest event (mandatory retirement regardless of visible damage)
Anti-contamination clothing	Torn, contaminated beyond decontamination limits, or exceeds single-use designation
Electrical gloves	Failed air test, past retest date, or visible cuts/punctures
Hard hat	Reached manufacturer service life, visible cracking, or exposure to significant impact

14.0 EMERGENCY PPE REQUIREMENTS

14.1 Fire Response

- Members of the site fire brigade don structural firefighting turnout gear meeting NFPA 1971.
- SCBA is required for any interior structural firefighting or entry into a smoke-charged atmosphere.

14.2 Chemical Release

- Initial responders not trained to HAZWOPER Technician level evacuate and establish a perimeter; no entry PPE is donned without HAZWOPER qualification.
- Qualified HazMat team members don Level A, B, or C protective ensembles per 29 CFR 1910.120, selected based on the substance and concentration involved.

14.3 Radiological Event

- Emergency responders don anti-contamination clothing and alarming dosimetry appropriate to declared area conditions before entry.
- SCBA is used in lieu of APR/PAPR when airborne radioactivity concentrations are unknown or exceed protection factor limits.

14.4 Confined Space Rescue

- Non-entry rescue is attempted first using a retrieval line and tripod/winch system per 29 CFR 1910.146.
- Entry rescue personnel don supplied-air respirators or SCBA and a full-body harness with retrieval line prior to entry.

15.0 HUMAN PERFORMANCE TOOLS

The following human performance tools are applied during PPE selection, donning, doffing, and use to reduce error and support safe task execution.

15.1 Self-Checking (STAR)

Individuals Stop, Think, Act, and Review before performing each critical PPE step, such as respirator donning or dosimeter placement, to confirm the correct action before proceeding.

15.2 Peer Checks

A second qualified individual independently verifies correct donning of anti-contamination clothing, respirators, and fall protection before high-risk entries.

15.3 Procedure Use and Adherence

Personnel use the current revision of this procedure and associated RWPs, following steps in sequence rather than from memory.

15.4 Place-Keeping

Checklists (Appendices A and B) are marked in real time as each PPE item is inspected or donned, preventing skipped or out-of-sequence steps.

15.5 Three-Way Communication

Dosimeter alarm set points, stay times, and RWP conditions are confirmed between the individual and Radiation Protection using send-receive-verify communication before entry.

15.6 Stop When Unsure

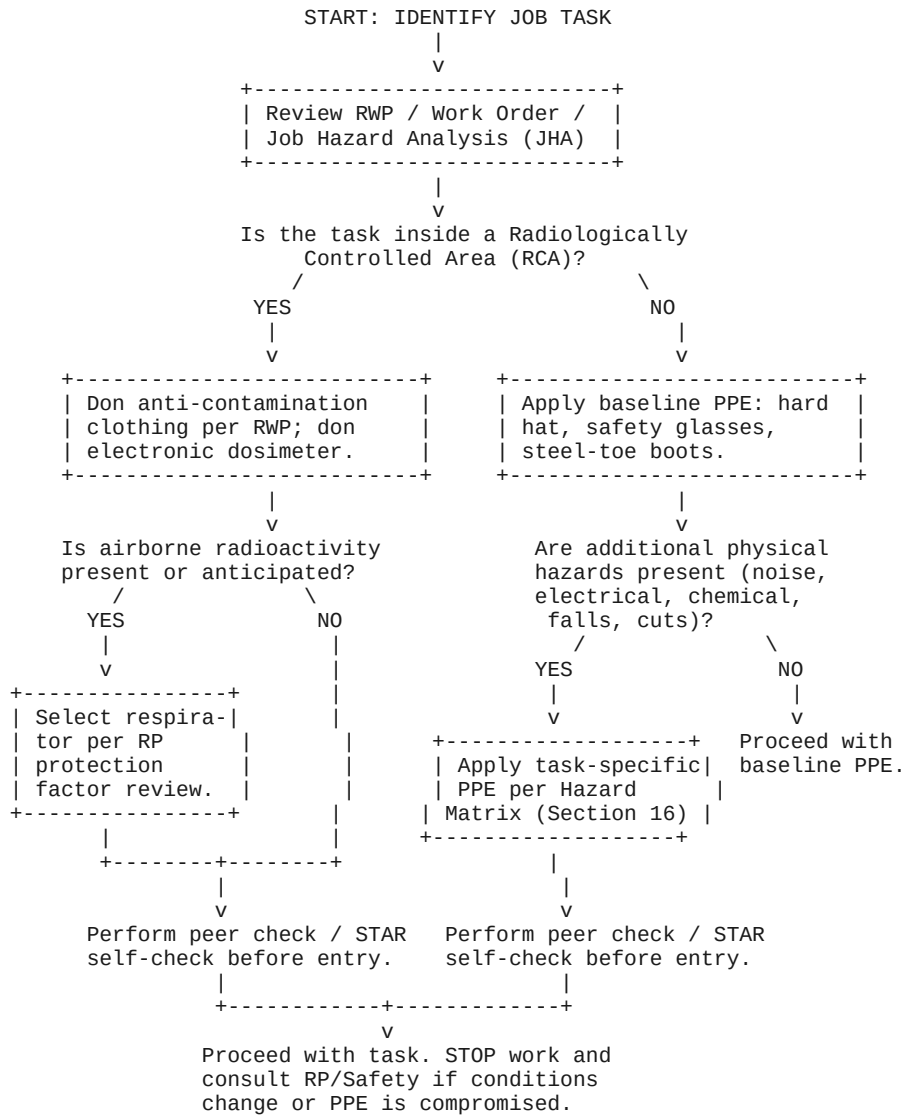
Any individual has the authority and obligation to stop work if PPE condition, fit, or adequacy for the hazard is in question, and to consult Radiation Protection or the Safety Department before proceeding.

16.0 HAZARD MATRIX

Hazard	Required PPE	Additional Controls	Required Training
Radiological contamination	Anti-contamination clothing, double gloves, shoe covers	RWP, contamination surveys, step-off pad	General Employee Radiation Training (GERT)
Airborne radioactivity	Respirator (APR/PAPR/SCBA per PF review)	Air sampling, engineering ventilation controls	Respiratory Protection Training, fit test
High noise (>85 dBA)	Hearing protection (single or dual)	Posted noise survey boundaries	Hearing Conservation Training
Electrical shock / arc flash	Arc-rated clothing, insulated gloves, face shield	Arc flash labeling, lockout/tagout	NFPA 70E Qualified Electrical Worker Training
Falling objects	Hard hat, steel-toe boots	Barricades, overhead work postings	General Safety Orientation
Chemical exposure	Chemical-resistant suit, gloves, goggles	SDS review, ventilation, spill kit availability	HAZCOM / SDS Training
Confined space atmosphere	Supplied-air respirator or SCBA, harness	Atmospheric monitoring, attendant, permit	Confined Space Entry Training
Falls from height	Full-body harness, shock-absorbing lanyard	Guardrails, anchor point verification	Fall Protection Training
Cuts / lacerations	Cut-resistant gloves	Tool guarding, job briefing	General Safety Orientation
Heat stress (anti-C use)	Cooling vest (as assigned)	Work/rest cycles, hydration, area monitoring	Heat Stress Awareness Training

17.0 PPE SELECTION FLOWCHART

The following decision tree summarizes the general PPE selection logic described in Sections 7 through 9. It does not replace the specific requirements of a governing RWP or JHA.



18.0 APPENDICES

Appendix A PPE Inspection Checklist

Item	Inspection Criteria Met?	Acceptable (Y/N)	Comments
Hard hat	Shell, suspension, service life		
Safety glasses	Lens condition, ANSI marking		
Hearing protection	Condition, fit		
Steel-toe boots	Sole, toe cap condition		
Gloves	Material, integrity		
Respirator	Seal, valves, cartridge date		
Fall protection harness	Webbing, hardware, stitching		
Electronic dosimeter	Power-on, zero reading, alarm test		

Appendix B Daily PPE Checklist

Date	Employee Name	Work Area	PPE Donned	RP Verification

Appendix C Respirator Fit Test Record

Employee Name	Respirator Make/Model	Test Method	Test Date	Pass/Fail	Next Due

Appendix D PPE Hazard Assessment Form

Task / Work Area	Hazard Identified	PPE Required	Assessed By	Date